

SEUNGPIIL LEE

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Personal Statement

Master’s researcher who moves in both directions between AI and the human mind: from AI back to the mind it imitates, and from the mind forward into the AI it shapes. Compares human and machine reasoning through cognitive science frameworks and mechanistic interpretability, with a current focus on memory-augmented agents that learn at test time. Draws on a background spanning literature, philosophy, and engineering to ask why models think the way they do.

EDUCATION

Gwangju Institute of Science and Technology (GIST) Master’s Program, Major in AI Convergence	<i>March 2025 – present</i>
Gwangju Institute of Science and Technology (GIST) B.S. in Electrical Engineering and Computer Science Minor in Mathematics, Minor in Literature	<i>March 2018 – December 2024</i> GPA: 3.66/4.50
UC Berkeley Berkeley Summer Session Program	<i>June 2019 – August 2019</i> GPA: 4.00/4.00

PUBLICATIONS

Journal Articles

- **Seungpil Lee***, Woochang Sim*, Donghyeon Shin*, Sanha Hwang, Wongyu Seo, Jiwon Park, Seokki Lee, Sejin Kim, and Sundong Kim. “Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus.” *ACM Transactions on Intelligent Systems and Technology* (**ACM TIST**), 2024.
- Sanha Hwang, **Seungpil Lee**, Sejin Kim, and Sundong Kim. “Solution Augmentation for ARC-AGI Problems Using GFlowNet: A Probabilistic Exploration Approach.” *Transactions on Machine Learning Research* (**TMLR**), 2025.

Conference Papers

- **Seungpil Lee***, Donghyeon Shin*, Klea Lena Kovacec, and Sundong Kim. “From Generation to Selection: Findings of Converting Analogical Problem-Solving into Multiple-Choice Questions.” *Findings of EMNLP* (**EMNLP Findings**), 2024.
- Hosung Lee*, Sejin Kim*, **Seungpil Lee**, Sanha Hwang, Jihwan Lee, Byung-Jun Lee, and Sundong Kim. “ARCLE: The Abstract and Reasoning Corpus Learning Environment for Reinforcement Learning.” *Conference on Lifelong Learning Agents (CoLLAs)*, 2024.

Preprints

- Heejun Kim*, **Seungpil Lee***, Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Taesup Kim, and Sundong Kim. “From Noise to Diversity: Random Embedding Injection in LLM Reasoning.” *arXiv preprint*, 2026.
- **Seungpil Lee**, Donghyeon Shin, Yunjeong Lee, and Sundong Kim. “Can Large Language Models Develop Gambling Addiction?” *arXiv preprint*, 2025.

Workshop Papers

- **Seungpil Lee***, Woochang Sim*, Donghyeon Shin*, Sejin Kim, and Sundong Kim. “Reasoning Abilities of Large Language Models through the Lens of Abstraction and Reasoning.” *NeurIPS Workshop on System-2 Reasoning at Scale*, 2024.
- Donghyeon Shin*, **Seungpil Lee***, Klea Lena Kovacec, and Sundong Kim. “Regulation Using Large Language Models to Generate Synthetic Data for Evaluating Analogical Ability.” *IJCAI Workshop on Analogical Abstraction*, 2024.

Domestic Publications

- **Seungpil Lee**, Jihwan Lee, and Sundong Kim. “Evaluating Prior Knowledge of ARC Using World Models.” *Korea Software Congress*, 2023.
- Jihwan Lee, **Seungpil Lee**, Sejin Kim, and Sundong Kim. “Extracting the Core Knowledge of ARC with the World Model.” *Korea Software Congress*, 2023.
- Donghyeon Shin, Sanha Hwang, Seokki Lee, Yunho Kim, **Seungpil Lee**, and Sundong Kim. “MC-LARC Benchmark to Measure LLM Reasoning Capability.” *Korea Software Congress*, 2023.

* indicates equal contribution

SKILLS

Programming	C, C++, Java, JavaScript, Python
Tools	LaTeX, Spring Boot, PyTorch, Hugging Face
Language	Korean (Native), English (Intermediate)

EXPERIENCE

Microsoft Research Asia (MSRA), Shanghai <i>Research Intern, Collaborative Research Program</i>	<i>March 2026 – August 2026</i>
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- Designing structured memory frameworks for self-evolving agents that update their knowledge at inference time without parameter retraining
- Studying how test-time memory generation enables compositional generalization on reasoning tasks where vanilla LLMs fail to reuse learned primitives
- Bridging mechanistic analysis of reasoning failures with memory-based mitigations

DataScience Lab, GIST <i>Undergraduate Intern → Master’s Student</i>	<i>September 2023 – present</i>
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- LLM evaluation with cognitive science tools (Language of Thought Hypothesis)
- LLM alignment using RLVR methods; Mechanistic interpretability with Sparse Autoencoders

Teaching Assistant, Computer Networking	<i>September 2023 – December 2025</i>
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- Designed assignments connecting theory to implementation; SDN demo with Ryu and Mininet

BioComputing Lab, GIST <i>Undergraduate Intern</i>	<i>June 2020 – January 2021</i>
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- Developed Spiking Neural Network methods with synthetic gradients

AWARDS & FUNDING

National Research Foundation of Korea (NRF) Funding Research funding for outstanding master’s students	<i>Sep 2025 – Sep 2026</i>
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Best Undergraduate Thesis Award, GIST EECS Awarded for graduation thesis on MC-LARC benchmark	<i>February 2025</i>
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Korean Government Scholarship, GIST College

Full scholarship for students at GIST

March 2018 – August 2024

Scholarship for Summer Session Abroad

UC Berkeley Summer Session scholarship

June – August 2019

Navy AI Competition

Award in drone object detection competition

August 2023

K-StartUp Finals

Advanced to finals as ML researcher and back-end developer

November 2023

EXTRA-CURRICULAR

Published Author – Short Story Collection

Published a 400-page short story collection exploring consciousness and identity

2019

AGIST – Deep Learning Study Group

Presentations on Spiking Neural Networks with STDP learning

August 2020 – January 2021